

Applying DAX Fundamentals in Power BI: From Measures to Contextual Analysis

Untitled - Power BI Desktop

Search

Sign in

File Home Insert Modeling View Optimize Help Table tools Measure tools

Share

Name Total Number of C...
Home table Orders
Format Decimal number
Data category Uncategorized
New measure Quick measure

Structure Formatting Properties Calculations

1 Total Number of Cookies Sold = SUM(Orders[Units Sold])

Visualizations

Build visual

Search

Filters on this page
Add data fields here

Filters on all pages
Add data fields here

Values
Add data fields here

Name [Total Number of Cookies Sold]

Cross-report Off
Keep all filters On

Data

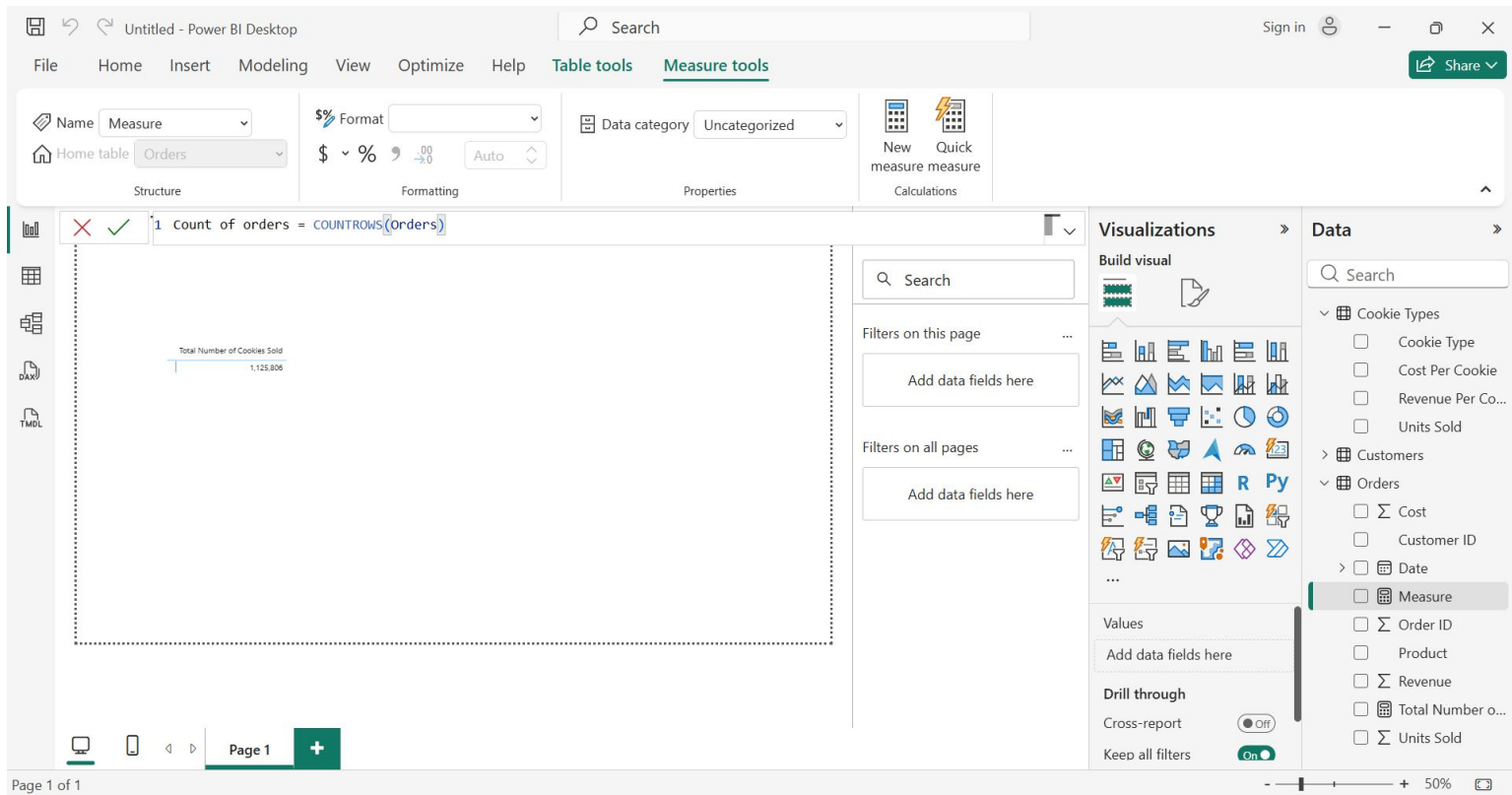
Search

- Cookie Types
 - ☐ Cookie Type
 - ☐ Cost Per Cookie
 - ☐ Revenue Per Co...
 - ☐ Units Sold
- Customers
- Orders
 - ☐ Cost
 - ☐ Customer ID
- Date
 - ☐ Order ID
 - ☐ Product
 - ☐ Revenue
 - ☐ Total Numbe...
 - ☐ Units Sold

Page 1

Page 1 of 1

50%



Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help Table tools Measure tools

Name: Measure Format: % Data category: Uncategorized

Home table: Orders

Structure Formatting Properties Calculations

1 Distinct Customer = DISTINCTCOUNT(Orders[Customer ID])

Write a DAX expression that calculates a value from your data.

Customer	Count
ACME Bites	206
Park & Shop Convenience Stores	114
Treat Delicious	92
Wholesome Foods	156
Total	700

Visualizations

Build visual

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Values

Add data fields here

Drill through

Cross-report

Keep all filters

Data

Customer ID

Name

Notes

Phone

State

Zip

Orders

Cost

Count of orders

Customer ID

Date

Measure

Order ID

Product

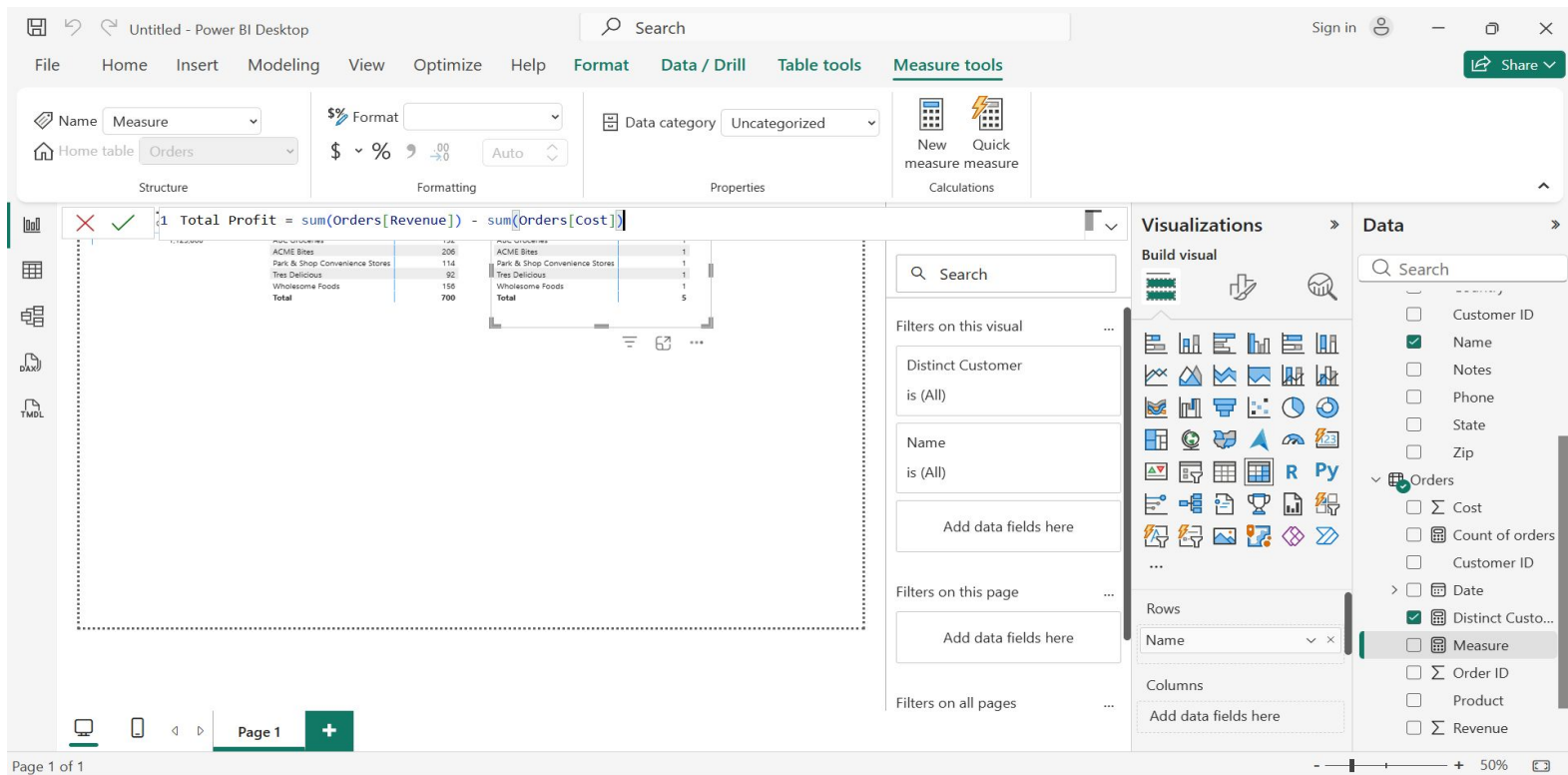
Revenue

Total Number ...

Page 1

Page 1 of 1

50%



Using Quick Measure:

Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools

Name: Sum of Revenue m... Format: General Data category: Uncategorized

Home table: Orders

Structure Formatting Properties

New Quick measure measure Calculations

1 Sum of Revenue minus Sum of Cost =
2 SUM('Orders'[Revenue]) - SUM('Orders'[Cost])

Name	Total Number of Cookies Sold	Name	Count of orders	Name	Count of orders
ABC Groceries	221023	ABC Groceries	132	ABC Groceries	132
ACME Bites	329677	ACME Bites	206	ACME Bites	206
Park & Shop Convenience Stores	179443	Park & Shop Convenience Stores	114	Park & Shop Convenience Stores	114
Tres Delicious	128769	Tres Delicious	92	Tres Delicious	92
Wholesome Foods	266890	Wholesome Foods	156	Wholesome Foods	156
Total	1125806	Total	700	Total	700

Name	Distinct Customer	Name	Distinct Customer	Total Profit	Sum of Revenue minus Sum of Cost
ABC Groceries	1	ABC Groceries	1	523,317.00	523,317.00
ACME Bites	1	ACME Bites	1	628,379.00	628,379.00
Park & Shop Convenience Stores	1	Park & Shop Convenience Stores	1	423,498.50	423,498.50
Tres Delicious	1	Tres Delicious	1	302,199.00	302,199.00
Wholesome Foods	1	Wholesome Foods	1	639,677.00	639,677.00
Total	5	Total	5	2,717,070.50	2,717,070.50

Visualizations

Build visual

Quick measure

Filters

Rows

Columns

Add data fields here

Search

- Phone
- State
- Zip
- Orders
 - Cost
 - Count of orders
 - Customer ID
 - Date
 - Distinct Custo...
 - Order ID
 - Product
 - Revenue
 - Sum of Rev...
 - Total Number ...
 - Total Profit
 - Units Sold

Page 1

Page 1 of 1

50%

Profit:

Untitled - Power BI Desktop

File Home Help **Table tools** Column tools

Share

Name: Profit

Data type: Decimal number

Format: General

Summarization: Sum

Data category: Uncategorized

Sort by column: Sort

Data groups: Groups

Manage relationships: Relationships

New column: Calculations

Structure

Formatting

Properties

1 Profit = Orders[Revenue] - Orders[Cost]

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976

Table: Orders (700 rows) Column: Profit (657 distinct values)

Data

- Cookie Types
 - Cookie Type
 - Cost Per Cookie
 - Revenue Per Cookie
 - Units Sold
- Customers
- Orders
 - Cost
 - Count of orders
 - Customer ID
- Date
 - Distinct Customer
 - Order ID
 - Product
- Profit
- Revenue

Profit Margin %:

Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools

Name Profit Margin % Format General Data category Uncategorized

Home table Orders \$ % % Auto

Structure Formatting Properties Calculations

1 Profit Margin % = [Total Profit] / SUM(Orders[Revenue])

Name	Total Number of Cookies Sold	Name	Count of orders	Name	Count of orders
ABC Groceries	221023	ABC Groceries	132	ABC Groceries	132
ACME Bites	323677	ACME Bites	206	ACME Bites	206
Park & Shop Convenience Stores	179445	Park & Shop Convenience Stores	114	Park & Shop Convenience Stores	114
Tres Delicious	128769	Tres Delicious	92	Tres Delicious	92
Wholesome Foods	266890	Wholesome Foods	156	Wholesome Foods	156
Total	1125866	Total	700	Total	700

Name	Distinct Customer	Name	Distinct Customer	Total Profit	Revenue of Minus Cost
ABC Groceries	1	ABC Groceries	1	\$23,317.00	\$23,317.00
ACME Bites	1	ACME Bites	1	\$28,379.00	\$28,379.00
Park & Shop Convenience Stores	1	Park & Shop Convenience Stores	1	\$23,498.50	\$23,498.50
Tres Delicious	1	Tres Delicious	1	\$32,199.00	\$32,199.00
Wholesome Foods	1	Wholesome Foods	1	\$39,877.00	\$39,877.00
Total	5	Total	5	\$217,070.50	\$217,070.50

Visualizations

Build visual

Quick measure

Filters

Rows

Name

Columns

Add data fields here

Data

Search

- ☐ Zip
- ☒ Orders
 - ☐ Σ Cost
 - ☐ Count of orders
 - ☐ Customer ID
 - ☐ Date
 - ☒ Distinct Customer
 - ☐ Σ Order ID
 - ☐ Product
 - ☐ Profit
 - ☐ Profit Margin %
 - ☐ Σ Revenue
 - ☒ Revenue of Minus Cost
 - ☐ Total Number of Cookies Sold
 - ☒ Total Profit
 - ☐ Σ Units Sold

Page 1 of 1

Page 1

50%

Cookie Types:

The screenshot displays the Microsoft Power BI Desktop application. The top ribbon includes tabs for File, Home, Insert, Modeling, View, Optimize, Help, Format, Data / Drill, Table tools, and Measure tools. The 'Measure tools' tab is active, showing options for Name (Measure), Format, Data category (Uncategorized), and buttons for New and Quick measure measure.

The main workspace shows a DAX measure formula in the formula bar:

```
1 Total Profit 2 = SUMX('Cookie Types','Cookie Types'[Units Sold]*('Cookie Types'[Revenue Per Cookie] - 'Cookie Types'[Cost Per Cookie]))
```

The formula bar also includes a 'Structure' section with 'Home table' set to 'Cookie Types' and a 'Formatting' section with currency symbols (\$, %, ,) and an 'Auto' button. The 'Properties' section shows 'Data category' as 'Uncategorized'.

The 'Visualizations' pane on the right shows a 'Build visual' section with various chart icons. The 'Data' pane on the right shows a search bar and a list of fields under 'Cookie Types':

- ☐ Cookie Type
- ☐ Cost Per Cookie
- ☒ Measure
- ☐ Revenue Per Co...
- ☐ Units Sold

Below these are sections for 'Columns' (Add data fields here), 'Drill through' (Cross-report: Off, Keep all filters: On), and a 'Share' button.

The bottom status bar shows 'Page 2 of 2' and a zoom level of 50%.

Time and date Function:

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Search

Sign in

Share

Name: Day of Week

Format: Whole number

Summarization: Sum

Data type: Whole number

Data category: Uncategorized

Sort by column

Data groups

Manage relationships

New column

Calculations

Structure

Formatting

Properties

Sort

Groups

Relationships

1 Day of week = WEEKDAY(Orders[Date],1)

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of Week
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1

Table: Orders (700 rows) Column: Day of Week (7 distinct values)

Data

Search

- > Cookie Types
- > Customers
- > Orders
 - Σ Cost
 - Count of orders
 - Customer ID
- > Date
 - Day of Week
 - Distinct Customer
 - Σ Order ID
 - Product
 - Profit
 - Profit Margin %
 - Σ Revenue
 - Revenue of Minus Cost
 - Total Number of Cookies Sold

Name: 'Orders'[Day of Week]

Find:

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Search

Sign in

Share

Name: Has Chocolate

Data type: Whole number

Format: Whole number

Summarization: Sum

Data category: Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Has Chocolate = FIND("Chocolate",Orders[Product],1,0)

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of Week	Has Chocolate
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7	1
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7	1
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2	1
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3	1
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6	1
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7	1
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2	1
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6	1
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3	1
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3	1
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2	1
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2	1
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4	1
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6	1
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4	1
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3	1
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3	1
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3	1
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3	1
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1	1

Data

Search

- > Cookie Types
- > Customers
- > Orders
 - Σ Cost
 - Count of orders
 - Customer ID
- > Date
 - Day of Week
 - Distinct Customer
 - Has Chocolate
 - Σ Order ID
 - Product
 - Profit
 - Profit Margin %
 - Σ Revenue
 - Revenue of Minus Cost

Table: Orders (700 rows) Column: Has Chocolate (3 distinct values)

If Statement:

Untitled - Power BI Desktop

File Home Help **Table tools** **Column tools**

Search

Sign in

Share

Name: Has Chocolate

Data type: Text

Format: Text

Summarization: Don't summarize

Data category: Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Has Chocolate = if(FIND("Chocolate",Orders[Product]),1,0)>0,"Has Chocolate","No Chocolate")

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of Week	Has Chocolate
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7	Has Chocolate
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7	Has Chocolate
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2	Has Chocolate
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3	Has Chocolate
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6	Has Chocolate
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7	Has Chocolate
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2	Has Chocolate
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6	Has Chocolate
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3	Has Chocolate
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3	Has Chocolate
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2	Has Chocolate
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2	Has Chocolate
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4	Has Chocolate
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6	Has Chocolate
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4	Has Chocolate
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3	Has Chocolate
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3	Has Chocolate
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3	Has Chocolate
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3	Has Chocolate
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1	Has Chocolate

Table: Orders (700 rows) Column: Has Chocolate (2 distinct values)

Data

Search

- Cookie Types
- Customers
- Orders
 - Cost
 - Count of orders
 - Customer ID
- Date
 - Day of Week
 - Distinct Customer
- Has Chocolate
- Order ID
- Product
- Profit
- Profit Margin %
- Revenue
- Revenue of Minus Cost

